

BROOME, Australia

WMO: 942030

Lat: 17.9475S

Lon: 122.2353E

Elev: 30

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	51.4	53.9	19.4	14.6	81.6	24.6	18.7	79.5	20.7	71.3	19.0	72.9	3.3	110	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.4	99.4	69.6	96.8	70.8	94.8	72.3	83.0	89.3	82.3	88.5	81.6	88.1	10.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.2	162.4	87.0	80.6	159.2	86.9	79.9	155.1	86.4	47.1	89.4	46.2	88.8	45.5	88.4	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.6	17.8	16.3	46.0	106.1	2.3	2.3	44.4	107.8	43.0	109.1	41.8	110.4	40.1	112.0		
(5)			40.5	85.0	1.6	1.5	39.4	86.1	38.5	87.0	37.6	87.8	36.4	88.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.1	85.2	84.7	85.1	83.3	76.7	72.1	71.1	72.2	77.0	82.2	85.3	86.2		
	DBStd	6.64	2.61	2.86	2.93	3.53	4.22	4.27	3.96	3.63	3.89	3.62	2.91	2.75		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	5	0	0	0	0	0	3	2	0	0	0	0	0		
	CDD50	10971	1091	971	1089	999	827	662	654	688	810	997	1060	1123		
	CDD65	5501	626	551	624	549	362	215	191	223	360	532	610	658		
	CDH74	66150	8193	7043	8106	6818	3781	1978	1869	2353	3503	5944	7773	8789		
	CDH80	30415	3940	3269	4022	3477	1650	644	594	873	1364	2446	3677	4459		
(14) Wind	WSAvg	8.6	10.0	9.3	8.1	7.0	7.2	7.9	7.5	7.5	8.2	9.3	10.1	10.6		
(15) Precipitation	PrecAvg	26.0	8.4	7.3	4.1	0.9	0.9	0.6	0.1	0.1	0.0	0.0	0.4	2.6		
	PrecMax	58.3	36.1	19.4	19.4	9.7	10.6	8.2	2.0	0.9	0.9	1.1	2.3	16.1		
	PrecMin	5.2	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	12.0	8.8	5.1	4.5	1.6	1.8	1.4	0.3	0.2	0.1	0.2	0.6	3.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.8	96.9	100.1	100.5	96.6	91.4	91.3	93.8	99.3	102.3	104.1	101.8		
		MCWB	77.9	75.5	72.8	69.9	66.3	65.8	62.9	61.6	62.7	66.2	68.3	72.0		
	2%	DB	93.2	93.3	96.6	96.9	93.4	88.2	88.4	90.6	95.3	97.0	96.9	95.6		
		MCWB	79.8	80.0	74.8	70.2	66.0	63.9	62.0	60.9	62.3	67.3	71.5	77.0		
	5%	DB	91.6	91.6	93.9	94.7	91.1	86.3	86.0	87.9	91.4	92.0	93.1	93.2		
		MCWB	80.3	80.5	77.7	70.6	66.2	63.4	61.5	60.1	62.9	71.5	75.5	78.6		
	10%	DB	90.3	90.1	91.7	92.5	88.6	84.0	83.4	85.0	86.9	89.3	91.0	91.5		
		MCWB	80.0	80.1	79.0	72.5	66.7	63.3	61.4	60.3	66.3	74.0	76.8	79.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.5	83.7	84.1	81.9	78.8	76.2	73.7	72.9	76.3	79.8	81.6	82.8		
		MCDB	90.4	89.6	90.2	87.9	83.9	81.5	78.1	78.4	81.0	85.6	89.4	89.9		
	2%	WB	82.4	82.9	82.9	80.9	76.8	73.5	71.3	71.6	75.1	79.0	80.6	81.8		
		MCDB	88.7	88.9	89.0	87.3	82.4	79.3	76.9	77.2	79.9	84.9	87.9	89.0		
	5%	WB	81.7	82.0	82.1	80.0	75.1	71.6	69.3	70.3	74.1	78.3	79.9	81.0		
		MCDB	88.1	88.2	88.2	86.7	81.8	77.7	76.3	76.4	79.7	84.3	87.1	88.4		
	10%	WB	81.0	81.3	81.3	79.1	73.4	69.5	67.5	69.0	73.1	77.5	79.2	80.4		
		MCDB	87.7	87.7	87.6	86.0	80.6	77.1	75.6	75.5	79.3	83.8	86.4	87.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	10.7	13.9	19.9	24.0	24.4	26.1	27.5	22.0	16.3	13.0	11.4		
		MCDBR	12.7	12.5	17.7	23.4	27.2	26.4	29.0	32.0	30.1	22.9	18.4	14.8		
		MCWBR	5.7	5.9	8.1	10.6	11.8	11.8	12.4	13.6	13.3	10.0	8.9	6.6		
	5% WB	MCDBR	11.8	11.1	13.9	16.2	20.2	20.6	22.8	23.3	19.1	14.8	13.2	12.3		
		MCWBR	6.1	6.0	7.1	7.9	10.2	10.4	11.3	11.7	9.3	6.5	6.4	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.422	0.413	0.390	0.370	0.355	0.331	0.328	0.336	0.367	0.405	0.409	0.409		
	taud		2.471	2.504	2.566	2.563	2.541	2.582	2.573	2.510	2.468	2.400	2.431	2.493		
	Ebn at Noon		293	293	292	287	280	282	286	293	295	292	295	297		
	Edh at Noon		38	36	33	31	30	27	28	32	36	40	39	37		
(44) All-Sky Solar Radiation	RadAvg		2082	2040	1962	1879	1620	1520	1613	1898	2177	2352	2433	2280		
	RadStd		195	228	208	89	102	97	88	54	55	66	68	136		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.12	N/A	N/A	+1.28	+0.70	+0.56	N/A	N/A	+414	+416
(47) Regional (3 neighbors)		+0.79	N/A	N/A	+0.97	+0.67	+0.58	N/A	N/A	+414	+416

Nomenclature: See separate page